# Pet chews (including dental chews): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

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PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified human-grade standard for pet chews (including dental chews), stated as sold. Pet food is legally feed; HMTc voluntarily certifies each analyte to the strictest human-food sovereign maximum for the nearest analog, with the laxer feed-law floor disclosed for contrast.

## 1 | How every number on this standard is set

Pet food is legally governed by feed law — EU Directive 2002/32/EC on undesirable substances in animal feed, FDA Center for Veterinary Medicine guidance, and AAFCO model regulations — whose maximum levels are generally laxer than the corresponding human-food limits. Japan (FAMIC pet-food standard) and China (draft pet-food guidance) set enforced or proposed pet-food maxima in the same lax range. The HMTc standard for each analyte is not the feed limit: it is the strictest applicable human-food sovereign maximum for the nearest human-food analog, adopted as a voluntary human-grade over-compliance. The feed floor is disclosed on every row and never selects the number.

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 20 | human-grade read-across (Hong Kong) |

| | | | |
|---|---|---|---|
| Cadmium (Cd) | Tier 1 | 40 | human-grade read-across (European Union) |
| Total arsenic (tAs) | Tier 2 | 100 | human-grade read-across (Paraguay) |
| Inorganic arsenic (iAs) | Tier 1 | 100 | human-grade read-across (Israel) |
| Mercury, total (tHg) | Tier 1 | 20 | human-grade read-across (Hong Kong) |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 800 | human-grade read-across (European Union) |
| Tin (Sn) | Tier 2 | 250,000 | human-grade read-across (Singapore) |
| Aluminium (Al) | Tier 2 | 100,000 | human-grade read-across (China) |
| Total chromium (Cr) | Tier 2 | 500 | human-grade read-across (Hong Kong) |
| Hexavalent chromium (Cr-VI) | Tier 2 | ≤ Cr | bounded by Cr (reflex speciation) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 20 | Hong Kong — Cereal-based food intended to be consumed principally by persons under the age of 36 months |
| Cadmium (Cd) | 40 | European Union — pulses, general |
| Total arsenic (tAs) | 100 | Paraguay — pulses, dried legume seeds, except soybean |
| Inorganic arsenic (iAs) | 100 | Israel — Dried legumes, including peanuts and soybeans |
| Mercury, total (tHg) | 20 | Hong Kong — Rice, husked rice, polished rice, maize, maize flour, wheat, wheat flour |

| | | |
|---|---|---|
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 800 | European Union — cereals except durum/rice/husked-rice/-pseudocereal/oat rows |
| Tin (Sn) | 250,000 | Singapore — human-food analog |
| Aluminium (Al) | 100,000 | China — human-food analog |
| Total chromium (Cr) | 500 | Hong Kong — Vegetables, other than pulses |
| Hexavalent chromium (Cr-VI) | ≤ Cr | Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell. |

## 4 | Cite this standard

Pendergrass, K. (2026). *Pet chews (including dental chews): HMTc standard*. Institute of Contaminant Standards.

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## References

[1] China — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Cd, Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Hong Kong — human-food maximum levels
Binds Pb, tHg, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] Israel — human-food maximum levels
Binds iAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[5] Paraguay — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[6] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

## STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

## DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/pet-chews-dental. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

## COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

## PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/pet-chews-dental.